

P3217R0

Adjoints to "Enabling list-initialization for algorithms": `find_last`

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Abstract

We extend the same changes that [\[P2248R8\]](#) applied to the rest of the find algorithms to `std::ranges::find_last`.

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§ 1. Changelog

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§ 2. Motivation and scope

In the Tokyo 2024 meeting [\[P2248R8\]](#) was adopted. Due to an oversight (which is entirely our fault) `std::ranges::find_last` was accidentally excluded from the algorithms for which a default template type parameter for the "value" argument was provided.

We propose to modify `find_last`'s specification, so that it matches the post-P2248 one for the rest of the algorithms (especially `find`).

§ 3. Proposed Wording

All the proposed changes are relative to [\[N4971\]](#), assuming that [\[P2248R8\]](#)'s wording has been merged already.

Modify `[alg.find.last]` as shown:

```
template<forward_iterator I, sentinel_for<I> S, class T, class Proj = identity>  
    requires indirect_binary_predicate<ranges::equal_to, projected<I, Proj>,  
        constexpr subrange<I> ranges::find_last(I first, S last, const T& value)  
  
template<forward_range R, class T, class Proj = identity, class T = projected_iterator_value_t<R, Proj>>  
    requires indirect_binary_predicate<ranges::equal_to, projected_iterator_value_t<R, Proj>,  
        constexpr borrowed_subrange_t<R> ranges::find_last(R&& r, const T& value)
```

§ 4. Acknowledgements

Thanks to Jens Maurer for pointing out this oversight of [\[P2248R8\]](#).

Thanks to KDAB for supporting this work.

All remaining errors are ours and ours only.

§ References

§ Informative References

[N4971]

Thomas Köppe. *Working Draft, Programming Languages — C++*. 18 December 2023. URL: <https://wg21.link/n4971>

[P2248R8]

Giuseppe D'Angelo. *Enabling list-initialization for algorithms*. URL: <https://isocpp.org/files/papers/P2248R8.html>